```sql
/* ============================================================
   RaceDay System - Database Creation Script
   Run on a clean SQL Server instance using SSMS.
   Matches the ERD: Role, User, Event, Category, EventEnrolment, Result
   ============================================================ */

IF DB_ID('RaceDayDB') IS NOT NULL
BEGIN
    ALTER DATABASE RaceDayDB SET SINGLE_USER WITH ROLLBACK IMMEDIATE;
    DROP DATABASE RaceDayDB;
END
GO

CREATE DATABASE RaceDayDB;
GO

USE RaceDayDB;
GO

/* ============================================================
   TABLE: Role
   Lookup table for role-based access (Organiser, Participant, Admin)
   ============================================================ */
CREATE TABLE Role (
    RoleID      INT IDENTITY(1,1) PRIMARY KEY,
    RoleName    NVARCHAR(50) NOT NULL UNIQUE
);
GO

/* ============================================================
   TABLE: [User]
   All people in the system. Role determines what they can do.
   ============================================================ */
CREATE TABLE [User] (
    UserID          INT IDENTITY(1,1) PRIMARY KEY,
    RoleID          INT NOT NULL,
    FullName        NVARCHAR(100) NOT NULL,
    Email           NVARCHAR(150) NOT NULL UNIQUE,
    PasswordHash    NVARCHAR(255) NOT NULL,
    CreatedAt       DATETIME NOT NULL DEFAULT GETDATE(),
    CONSTRAINT FK_User_Role FOREIGN KEY (RoleID) REFERENCES Role(RoleID)
);
GO

/* ============================================================
   TABLE: Event
   Created by a User whose role is Organiser
   ============================================================ */
CREATE TABLE Event (
    EventID         INT IDENTITY(1,1) PRIMARY KEY,
    OrganiserID     INT NOT NULL,
    Name            NVARCHAR(150) NOT NULL,
    EventDate       DATE NOT NULL,
    Location        NVARCHAR(200) NOT NULL,
    Description     NVARCHAR(1000) NULL,
    CONSTRAINT FK_Event_Organiser FOREIGN KEY (OrganiserID) REFERENCES [User](UserID)
);
GO

/* ============================================================
   TABLE: Category
   Belongs to an Event (e.g. "10km", "21km")
   ============================================================ */
CREATE TABLE Category (
    CategoryID      INT IDENTITY(1,1) PRIMARY KEY,
    EventID         INT NOT NULL,
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    Name            NVARCHAR(100) NOT NULL,
    Distance        DECIMAL(6,2) NOT NULL,
    Fee             DECIMAL(8,2) NOT NULL DEFAULT 0,
    CONSTRAINT FK_Category_Event FOREIGN KEY (EventID) REFERENCES Event(EventID)
);
GO

/* ============================================================
   TABLE: EventEnrolment
   Junction table: a Participant (User) enrolling in a Category
   ============================================================ */
CREATE TABLE EventEnrolment (
    EnrolmentID     INT IDENTITY(1,1) PRIMARY KEY,
    ParticipantID   INT NOT NULL,
    CategoryID      INT NOT NULL,
    EnrolmentDate   DATETIME NOT NULL DEFAULT GETDATE(),
    Status          NVARCHAR(30) NOT NULL DEFAULT 'Confirmed',
    CONSTRAINT FK_Enrolment_Participant FOREIGN KEY (ParticipantID) REFERENCES
[User](UserID),
    CONSTRAINT FK_Enrolment_Category FOREIGN KEY (CategoryID) REFERENCES
Category(CategoryID),
    CONSTRAINT UQ_Participant_Category UNIQUE (ParticipantID, CategoryID)
);
GO

/* ============================================================
   TABLE: Result
   One result per enrolment (1:1)
   ============================================================ */
CREATE TABLE Result (
    ResultID        INT IDENTITY(1,1) PRIMARY KEY,
    EnrolmentID     INT NOT NULL UNIQUE,
    FinishTime      TIME NULL,
    Position        INT NULL,
    Status          NVARCHAR(30) NOT NULL DEFAULT 'Finished',
    CONSTRAINT FK_Result_Enrolment FOREIGN KEY (EnrolmentID) REFERENCES
EventEnrolment(EnrolmentID)
);
GO

/* ============================================================
   SEED DATA
   ============================================================ */

-- Roles
INSERT INTO Role (RoleName) VALUES ('Organiser'), ('Participant'), ('Admin');
GO

-- Users: 2 Organisers, 2 Participants
INSERT INTO [User] (RoleID, FullName, Email, PasswordHash) VALUES
((SELECT RoleID FROM Role WHERE RoleName = 'Organiser'),   'Thabo Mokoena',
'thabo.mokoena@raceday.com',  'hashed_password_1'),
((SELECT RoleID FROM Role WHERE RoleName = 'Organiser'),   'Lindiwe Nkosi',
'lindiwe.nkosi@raceday.com',  'hashed_password_2'),
((SELECT RoleID FROM Role WHERE RoleName = 'Participant'), 'Sipho Dlamini',
'sipho.dlamini@example.com',  'hashed_password_3'),
((SELECT RoleID FROM Role WHERE RoleName = 'Participant'), 'Anele Botha',
'anele.botha@example.com',    'hashed_password_4');
GO

-- Events: 3 events, each created by an Organiser
INSERT INTO Event (OrganiserID, Name, EventDate, Location, Description) VALUES
((SELECT UserID FROM [User] WHERE Email = 'thabo.mokoena@raceday.com'), 'Johannesburg
City Run',   '2026-10-10', 'Johannesburg, GP', 'Annual road race through the city
centre.'),
```

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((SELECT UserID FROM [User] WHERE Email = 'thabo.mokoena@raceday.com'), 'Sandton
Night Race',      '2026-11-21', 'Sandton, GP',      'Evening race with festival
atmosphere.'),
((SELECT UserID FROM [User] WHERE Email = 'lindiwe.nkosi@raceday.com'), 'Soweto
Heritage Marathon', '2026-12-05', 'Soweto, GP',       'Marathon celebrating local
heritage routes.');
GO

-- Categories: at least one per event
INSERT INTO Category (EventID, Name, Distance, Fee) VALUES
((SELECT EventID FROM Event WHERE Name = 'Johannesburg City Run'),    '5km',  5.0,
100.00),
((SELECT EventID FROM Event WHERE Name = 'Johannesburg City Run'),    '10km', 10.0,
150.00),
((SELECT EventID FROM Event WHERE Name = 'Sandton Night Race'),       '10km', 10.0,
180.00),
((SELECT EventID FROM Event WHERE Name = 'Soweto Heritage Marathon'), '21km', 21.1,
250.00),
((SELECT EventID FROM Event WHERE Name = 'Soweto Heritage Marathon'), '42km', 42.2,
350.00);
GO

-- Event enrolments: sample participants enrolling in categories
INSERT INTO EventEnrolment (ParticipantID, CategoryID, Status) VALUES
((SELECT UserID FROM [User] WHERE Email = 'sipho.dlamini@example.com'),
 (SELECT CategoryID FROM Category WHERE Name = '10km' AND EventID = (SELECT EventID
FROM Event WHERE Name = 'Johannesburg City Run')),
 'Confirmed'),
((SELECT UserID FROM [User] WHERE Email = 'anele.botha@example.com'),
 (SELECT CategoryID FROM Category WHERE Name = '5km' AND EventID = (SELECT EventID
FROM Event WHERE Name = 'Johannesburg City Run')),
 'Confirmed'),
((SELECT UserID FROM [User] WHERE Email = 'anele.botha@example.com'),
 (SELECT CategoryID FROM Category WHERE Name = '21km'),
 'Confirmed');
GO

-- Results: sample results for completed enrolments
INSERT INTO Result (EnrolmentID, FinishTime, Position, Status) VALUES
((SELECT EnrolmentID FROM EventEnrolment WHERE ParticipantID = (SELECT UserID FROM
[User] WHERE Email = 'sipho.dlamini@example.com')
    AND CategoryID = (SELECT CategoryID FROM Category WHERE Name = '10km' AND EventID
= (SELECT EventID FROM Event WHERE Name = 'Johannesburg City Run'))),
 '00:48:32', 1, 'Finished');
GO
```